Mark Dostalík

mark.dostalik@gmail.com | mark-dostalik.github.io | mark-dostalik | mark-dostalik | Mark Dostalík

Summary

Machine learning engineer with 5+ years of industry experience in computer vision, from satellite and biomedical imaging to automotive. Former tech lead of a 4-engineer team at MSD building an image-analysis package used daily by ~20 scientists. Ph.D. in mathematical modeling with 9 peer-reviewed publications.

Experience

Filuta AI — *Prague, Czechia*

Senior Machine Learning Engineer — *Jul. 2026 – present*

- Develop models matching infotainment screens across test runs in different languages for automated UI testing.
- Build object detection and OCR for automotive cockpit software; track experiments in MLflow and deploy with Docker on Azure.
- *Stack:* Python, PyTorch, Pydantic, PyTest, MLflow, Azure, Docker, GitHub Actions.

MSD Czech Republic — *Prague, Czechia*

Senior Data Scientist / Tech Lead — *Jul. 2022 – Jun. 2026*

- Led a team of 4 engineers building a Python package for high-resolution biological image analysis, used daily by ~20 scientists.
- Designed segmentation, image registration and statistical analysis pipelines scaling to hundreds of GB of imaging data.
- Introduced automated testing, CI/CD (GitHub Actions) and packaged releases to the internal Artifactory.
- Set up the AWS infrastructure (EC2, S3) for running image-processing workloads in the cloud.
- *Stack:* Python, PyTorch, torchvision, OpenCV, scikit-image, scikit-learn, NumPy, SciPy, Polars, Pydantic, PyTest, AWS.

Mathematical Institute of Charles University — *Prague, Czechia*

Postdoctoral Researcher — *Oct. 2021 – Jun. 2022*

- Developed a mathematical model of cell membrane growth combining continuum mechanics and molecular dynamics.
- Solved the resulting systems of partial differential equations numerically with finite element methods (Python, FEniCS).
- Co-developed OrganL, a dynamic triangulation method for biomembranes, published in *Biophysical Journal* (2024).

Spaceknow — *Prague, Czechia*

Machine Learning Engineer — *Nov. 2016 – Sep. 2017*

- Trained convolutional neural networks for semantic segmentation of buildings in satellite imagery.
- Ran ML pipelines both on premises and on Google Cloud Platform.
- *Stack:* Python, TensorFlow, Keras, scikit-learn.

Education

Faculty of Mathematics and Physics, Charles University — *Prague, Czechia*

Ph.D. in Mathematical and Computer Modeling — *Oct. 2017 – Sep. 2021*

- Focus: stability of dynamical systems, partial differential equations, continuum thermodynamics.
- Published 9 peer-reviewed papers (8 journal, 1 proceedings), full list on Google Scholar.
- Led a Charles University Grant Agency project as principal investigator (2019–2021), with 4 published results.

Skills

Languages	Python (OOP, typed, tested), Bash, Rust (basics)
ML & CV	PyTorch, torchvision, TensorFlow/Keras, scikit-learn, OpenCV, scikit-image
Data	NumPy, SciPy, Polars, Pandas, Plotly, Matplotlib
Engineering	Pydantic, PyTest, Git, Docker, CI/CD (GitHub Actions), Claude Code
Cloud & MLOps	AWS (EC2, S3, EFS, Lambda), Microsoft Azure, Google Cloud Platform, MLflow